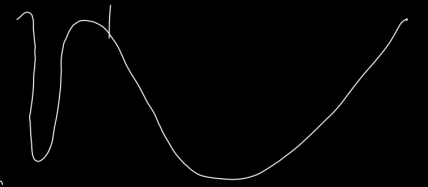


Optimization in NN



$$F(w) = \frac{1}{N} \sum_{i=1}^N \ell_i(w) \rightarrow \min_w, \quad N \gg 1$$

$$\text{SGD: } \begin{cases} I_k \subset \text{Unif}(1, 2, \dots, N) \end{cases}$$

$$\begin{cases} g_k = \frac{1}{|I_k|} \sum_{i \in I_k} \nabla \ell_i(w_k) \end{cases}$$

$$\text{Var}(\varepsilon_k) \xrightarrow{|I_k| \rightarrow N} 0$$

$$\begin{cases} w_{k+1} = w_k - \alpha_k g_k = w_k - \alpha_k (\nabla F(w_k) + \varepsilon_k) \end{cases}$$

$$f(w) \rightarrow \min_w$$



$$GD: w_{k+1} = w_k - \alpha_k \nabla f(w_k)$$

$$Newton: w_{k+1} = w_k - \alpha_k \left(\nabla^2 f(w_k) \right)^{-1} \nabla f(w_k)$$

curvature scaling matrix

$$L-BFGS: w_{k+1} = w_k - \alpha_k L-BFGS(\nabla f(w_k), \{s_i, y_i\}_{i=k-m}^{k-1})$$

$$s_i = w_{i+1} - w_i, y_i = \nabla f(w_{i+1}) - \nabla f(w_i)$$

SGD + momentum

$$w_{k+1} = w_k - \alpha_k m_k$$

$$m_k = \beta \cdot m_{k-1} + g_k, \quad \beta \in (0, 1)$$

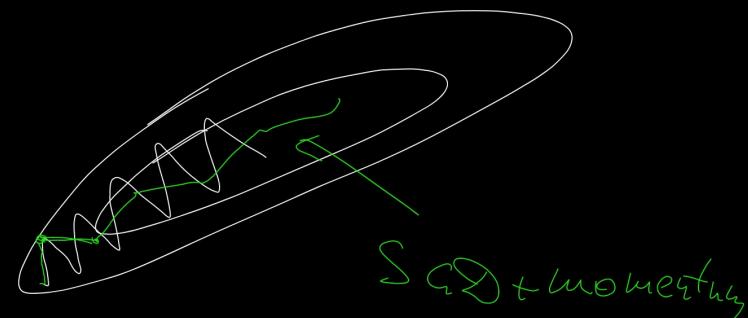
$$m_0 = g_0$$

$$m_1 = \beta m_0 + g_1 = \beta g_0 + g_1$$

$$m_2 = \beta^2 g_0 + \beta g_1 + g_2$$

⋮

$$m_k = \sum_{i=0}^k \beta^{k-i} g_i = \text{EMA}(\{g_k\})$$

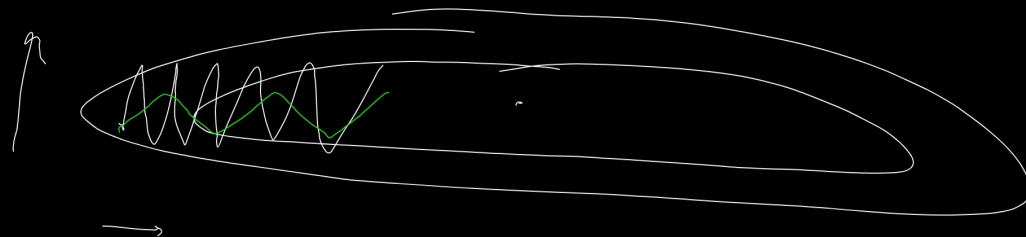


Weight Decay

$$F(\omega) = \frac{1}{N} \sum_i f_i(\omega) + \frac{\lambda}{2} \|\omega\|_2^2 \rightarrow \min_{\omega}$$

$$\text{SGDW: } \omega_{k+1} = \omega_k - \alpha_k (M_k + \lambda \omega_k)$$

RMSprop



RMSprop = diag. approx. of curv. matrix

$$\omega_{k+1,i} = \omega_{k,i} - \alpha_k \left(\frac{g_{k,i}}{\sqrt{\delta_{k,i} + \epsilon}} + \lambda \omega_{k,i} \right)$$

$$\delta_{k,i} = \beta \delta_{k-1,i} + (1-\beta) g_{k,i}^2$$

$$(1-\beta) \sum_{i=0}^k \beta^{k-i} = (1-\beta) \frac{1-\beta^k}{1-\beta} = 1-\beta^k$$

ADAM W

ADAM = RMSprop + momentum

$$\bar{w}_{k+1,i} = \bar{w}_{k,i} - \alpha_k \left(\frac{\hat{m}_{k,i}}{\sqrt{\hat{v}_{k,i} + \epsilon}} + \lambda \bar{w}_{k,i} \right)$$

$$\hat{m}_{k,i} = \frac{m_{k,i}}{1-\beta_1^k}$$

$$\hat{v}_{k,i} = \frac{v_{k,i}}{1-\beta_2^k}$$

$$m_{k,i} = \beta_1 m_{k-1,i} + (1-\beta_1) g_{k,i}$$

$$v_{k,i} = \beta_2 v_{k-1,i} + (1-\beta_2) g_{k,i}^2$$

$$m_0 = (1-\beta) g_0$$

$$m_1 = \beta(1-\beta) g_0 + (1-\beta) g_1$$

$$m_2 = \beta^2(1-\beta) g_0 + \beta(1-\beta) g_1 + (1-\beta) g_2$$

$$m_k = (1-\beta) \sum_{i=0}^k \beta^{k-i} g_i$$

$$E m_k = (1-\beta) \sum_{i=0}^k \beta^{k-i} \underbrace{E g_i}_{\text{constant}}$$

AMS Grad : $\hat{v}_{k,i} = \max(\hat{v}_{k-1,i}, \beta_2 \hat{v}_{k,i} + (1-\beta_2) g_{k,i}^2)$

Step size recommendation:

SGD $\alpha_k = 10^{-2}, 10^{-3}$

ADAM $\alpha_k = 10^{-4}, 10^{-5}$

More advanced curvature matrix approximations:

Shampoo, SOAP

MuOn (Momentum Orthogonalization)

W - matrix

In classic GD: $f(\omega_k + d) = f(\omega_k) + \nabla f(\omega_k)^T d +$

$$\min_{d: \|d\|_2 \leq 1} \nabla f(\omega_k)^T d \Rightarrow d_{\text{opt}} = - \frac{\nabla f(\omega_k)}{\|\nabla f(\omega_k)\|_2} + \mathcal{O}(\|d\|)$$

$$\text{GD: } \omega_{k+1} = \omega_k - \alpha_k \nabla f(\omega_k)$$

$$\langle \nabla f(w_k), \mathcal{D} \rangle$$

$$f(w_k + \mathcal{D}) = f(w_k) + t_2(\nabla f(w_k)^T \mathcal{D}) + o(\|\mathcal{D}\|)$$

$$\min_{\mathcal{D}: \|\mathcal{D}\|_F \leq 1} t_2(\nabla f(w_k)^T \mathcal{D}) \Rightarrow \mathcal{D}_{\text{opt}} = - \frac{\nabla f(w_k)}{\|\nabla f(w_k)\|_F}$$

Min On

$$\min_{\mathcal{D}: \|\mathcal{D}\|_2 \leq 1} t_2(\nabla f(w_k)^T \mathcal{D})$$

SV $\mathcal{D}$ decomp.

$$\nabla f(w_k) = U \Sigma V^T$$

$$\mathcal{D}_{\text{opt}} = U V^T$$

$$\sigma_i(\mathcal{D}_{\text{opt}}) = 1$$

$$\|\mathcal{D}\|_2 = \sigma_{\max}(\mathcal{D})$$

$$\sigma_i(\mathcal{D}) = \sqrt{\lambda_i(\mathcal{D}^T \mathcal{D})}$$

→ Singular value

→ Eigenvalue

$$y = Wx, \quad \|y\|_2 = \|Wx\|_2 \leq \|W\|_2 \cdot \|x\|_2$$

$$\frac{\sigma_{\max}(\nabla f(w_k))}{\sigma_{\min}(\nabla f(w_k))} \gg 1$$

$$W = U \Sigma V^T$$

$$y = U \Sigma \underbrace{V^T x}_{\text{rotation}}$$

Space stretching

$$X = \text{Newton-Schulz}(M), \quad X \approx UV^T$$

$$\text{Init. } X_0 = \frac{M}{\|M\|_F + \epsilon}, \quad M = U \Sigma V^T$$

for $k=0, 1, 2, \dots$ #iters:

$$A_k = X_k X_k^T$$

$$X_{k+1} = a X_k + (\beta A_k + c A_k^2) X_k$$

$$a = 3.44, \beta = \dots, c = \dots$$

MuOn

$$M_k = \nabla f(W_k)$$

$$Q_k = NS(M_k)$$

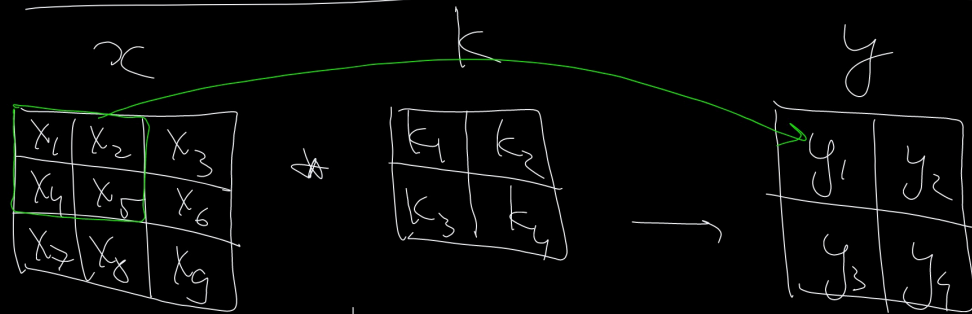
$$W_{k+1} = W_k - \alpha_k (Q_k + \lambda W_k)$$

Two groups of parameters:

MuOn: W in intermediate linear layers

AdamW: σ, σ in Batch Norm, biases b , parameters of the first and 'last layer', ...

Convolution operation

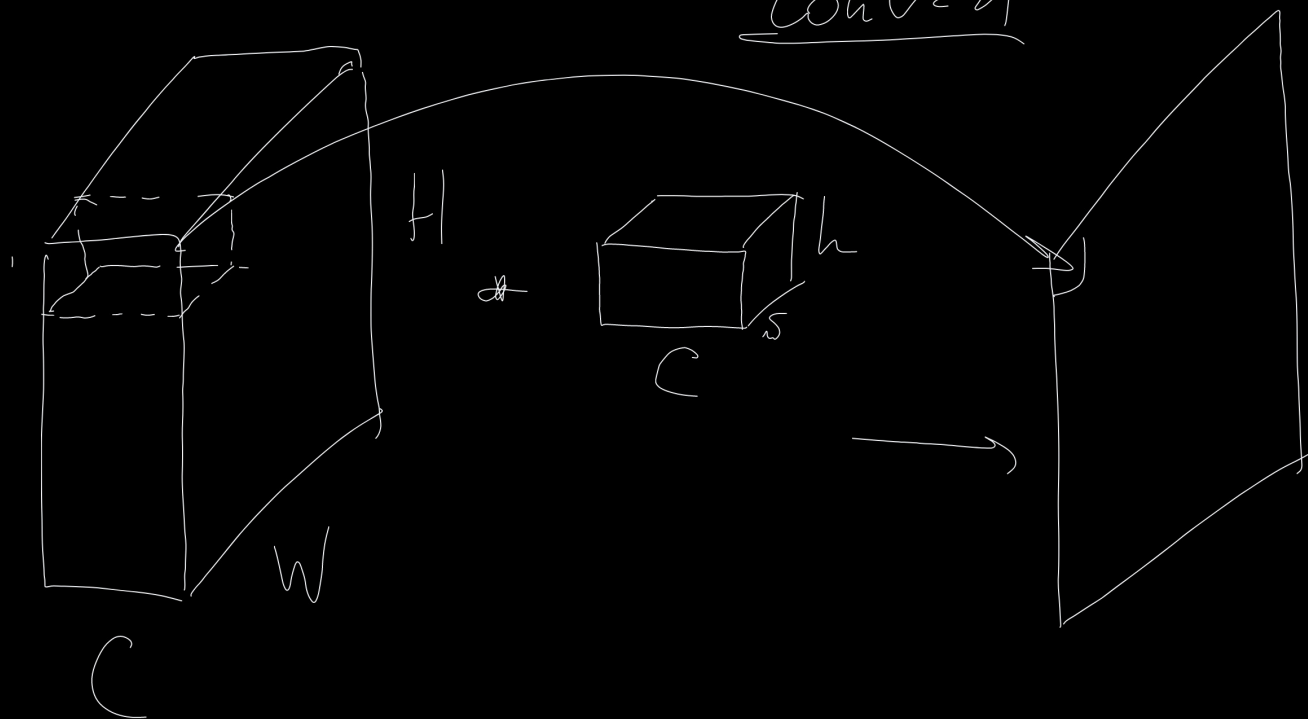


$$y_{ij} = \sum_{p,q=0}^{w-1, h-1} k_{pq} x_{i+p, j+q}$$

$$y = x * k \Leftrightarrow \text{vec}(y) = K_{\text{conv}} \text{vec}(x)$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} k_1 & k_2 & 0 & k_3 & k_4 & 0 & 0 & 0 \\ 0 & k_1 & k_2 & 0 & k_3 & k_4 & 0 & 0 \\ 0 & 0 & 0 & k_1 & k_2 & 0 & k_3 & k_4 \\ 0 & 0 & 0 & 0 & k_1 & k_2 & 0 & k_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_9 \end{bmatrix}$$

Conv2d



$$y = x + k$$

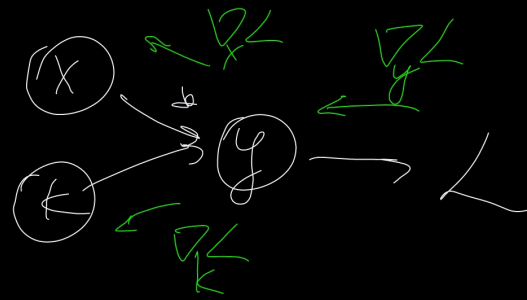
$$y_{ij} = \sum_{p,q} k_{pq} x_{i+p,j+q}$$

$$dL = \sum_{i,j} \frac{\partial L}{\partial y_{ij}} dy_{ij} = \sum_{i,j} \frac{\partial L}{\partial y_{ij}} \sum_{p,q} (dk_{pq} x_{i+p,j+q} +$$

$$+ k_{pq} dx_{i+p,j+q}) = \sum_{p,q} dk_{pq} \left(\sum_{i,j} \frac{\partial L}{\partial y_{ij}} x_{i+p,j+q} \right) +$$

$$+ \sum_{u,v} dx_{uv} \left(\sum_{p,q} k_{pq} \frac{\partial L}{\partial y_{u-p,v-q}} \right)$$

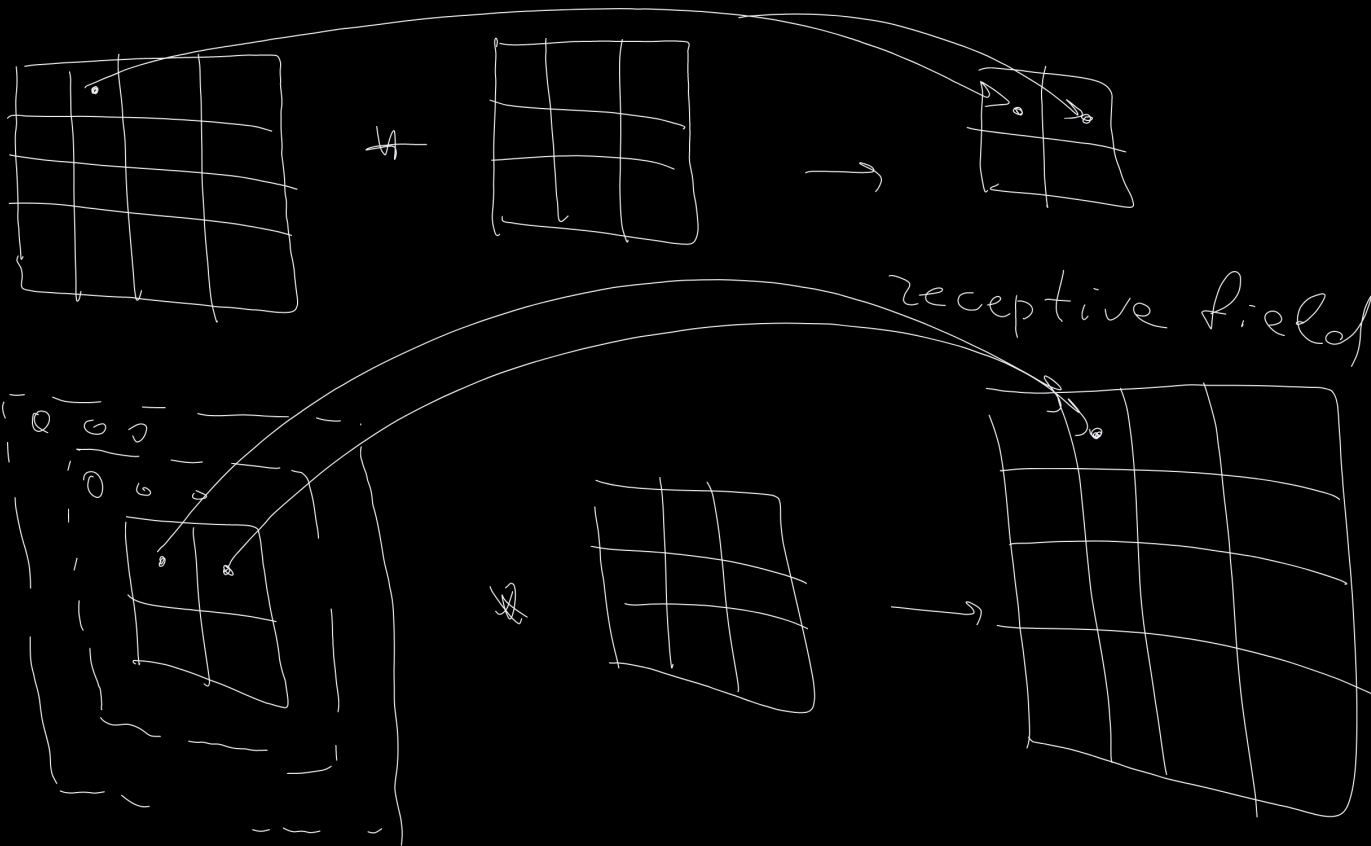
$u = i+p$
 $v = j+q$
 $\frac{\partial L}{\partial x_{uv}}$



$$\frac{\partial L}{\partial k} = \frac{\partial L}{\partial y}$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} + \text{reshape}(k)$$

Receptive field



$$112 \times 112 \times 128 \times 128 \times 3 \times 3 = \\ = 56 \times 56 \times 256 \times 256 \times 3 \times 3$$

Conv 2d

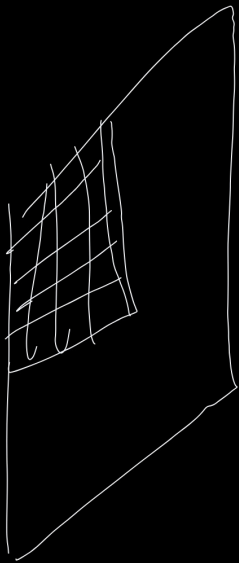
input $X \in \mathbb{R}^{H \times W \times C_{in}}$

$$K \in \mathbb{R}^{h \times w \times C_{in} \times C_{out}}$$

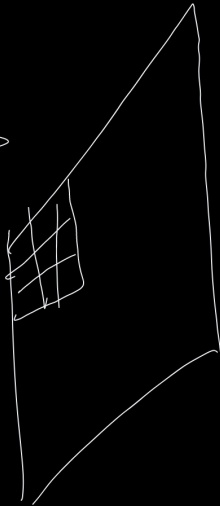
$$Y = X * K, Y \in \mathbb{R}^{H_{out} \times W_{out} \times C_{out}}$$

Computational
complexity:

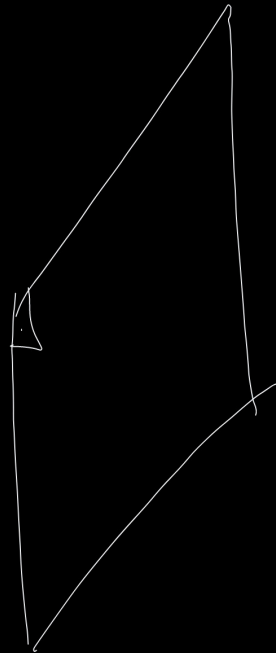
$$O(H_{out} W_{out} C_{out} \times h \cdot w \cdot C_{in})$$

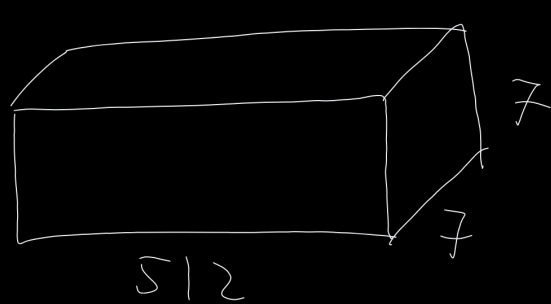


conv 3×3



conv
 3×3





Avg Pool
→

